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Vapor Stone Rail Systems

Generic Software Documentation Project

Project # 7260

Portable Test Application

Software Requirements Specifications (SRS)

Contract Number		CDRL Number	
		Customer Ref. Number	

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Preface

This Software Requirements Specification (SRS) complies with the requirements of the IEEE 830-1998, IEEE Recommended Practice for Software Requirements Specifications, approved on 1998-06-25 and IEEE Std 1558-2004, IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems approved on 2005-05-23.

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1 INTRODUCTION

1.1 Purpose

This document lists the specifications required for the Door Module System (DMS) VAP4 Portable Test Application(PTA) developed by Vapor Stone Rail Systems.

The intended audience is the VSRS software development and maintenance team members, employees of transit authorities and car builders using this software.

1.2 Scope

The scope of this Software Requirement Specification covers the Door Module System Portable Test Application software interface and functionalities.

The Door Module System Portable Test Application software will interface with the Door Module System application via hardware interface(s) (HW_COM) . The Portable Test Application software will be able to retrieve the Door Module System log data, to do real time monitoring to the Door Module System, to change door parameters, and to perform Door Module System software upload. It will provide testing, analysis, and troubleshooting of the Door Module System.

This document specifies the generic part of the Portable Test Application Software Requirements Specifications. Specific project's configuration information is documented into the Software Applications Configuration Specifications (SACS) which is specific for each project.

1.3 Definitions, acronyms, and abbreviations

Definitions and acronyms listed below establish meaning in the context of this Software Requirements Specifications.

1.3.1 Definitions

Boot Loader:	Software starting the target application of a door module.
Car Network:	Data bus connecting the Door Control Module with the rest of the train (others Door Control Modules in the car and others cars).
Communication Gateway Module:	A specific processor based electronic module running a Communication Gateway Module application. It acts as gateway between the Door Control Module application and the Car Network/Portable Test Application.
Configuration File:	A file within the Door Control Module containing all the configurable parameters value.
Dongle:	A hardware device attached to a computer port to enable authorization of certain software functions.
Door Control Module (DCM):	A specific processor based electronic module running a Door Control Module application. It is used for controlling the door motor.
Door Control Unit (DCU):	The final unit that is used to control one or more door panels. This unit is normally made of one or two door control modules running the Door Control Module application and possibly one Communication Gateway Module application.

Door Module System (DMS):	A term that is used for generically identifying a processor based electronic module running any software within any Door Control Unit.
Graphical User Interface (GUI):	Interface used for the communication between the User and the Portable Test Application software. This is the visible part of the Portable Test Application
HW_COM:	Communication interface used for communication between Portable Test Application and a door module application, which can be USB, serial or Ethernet.
User:	A human operating the Portable Test Application, issuing request and reading results

1.3.2 Acronyms

API	Application Program Interface
BLA	Boot Loader Application
CPU	Central Process Unit
DCM	Door Control Module
DMS	Door Module System
DMSA	Door Module System Application
EDF	Event Description File
FT	File Transfer
GUI	Graphical User Interface
HD	Hard Disk
HW	Hardware
ICD	Interface Control Document
IEEE	Institute of Electrical and Electronics Engineers
IO	Input and Output
LED	Light-Emitting Diode
MDC	Master Door Controller
OPDF	Operating Parameters Description File
OS	Operating System
PC	Personal Computer
PTA	Portable Test Application
RTDM	Real-Time Data Monitoring
SACS	Software Applications Configuration Specifications
SCI	Software Configuration Item
SFD	System Functional Description
SRS	Software Requirements Specifications
USB	Universal Serial Bus
VSRS	Vapor Stone Rail Systems

1.4 References

Standards

IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specifications, 1998-06-25, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Recommended Practice for Software Documentation for Rail Equipment and Systems, 2005-05-23, IEEE, www.ieee.org

VSRS documents

CMM-01	Software Engineering Process, VSRS, Rev. 3, 2001-03-31
CMMI-02	Software engineering policy manual, VSRS, Rev. 1, 2008-08-17
S17-01	Software Requirements Specification (SRS) Template, Rev. 0, 2002-01-15
SACS	Software Applications Configuration Specifications, Project's specific.
200310616	Software User Document, Rev. B, 2011-03-01.
201111489	Portable Test Application Interface Control Document (ICD), VSRS, Rev 0.1

1.5 Overview

This document describes PTA software inputs, outputs and functions. The SRS specifies the performance requirements and constraints imposed on the design of the Door Module System Portable Test Application.

There are three sections in this document:

- Section 1- *Introduction, Definitions and Overview*
- Section 2- *Overall Description*
- Section 3- *Specific Requirements*

2 OVERALL DESCRIPTION

2.1 Product perspective

The PTA software is designed for the DMS to help the maintenance staff in maintaining, inspecting, troubleshooting, adjusting, testing and repairing.

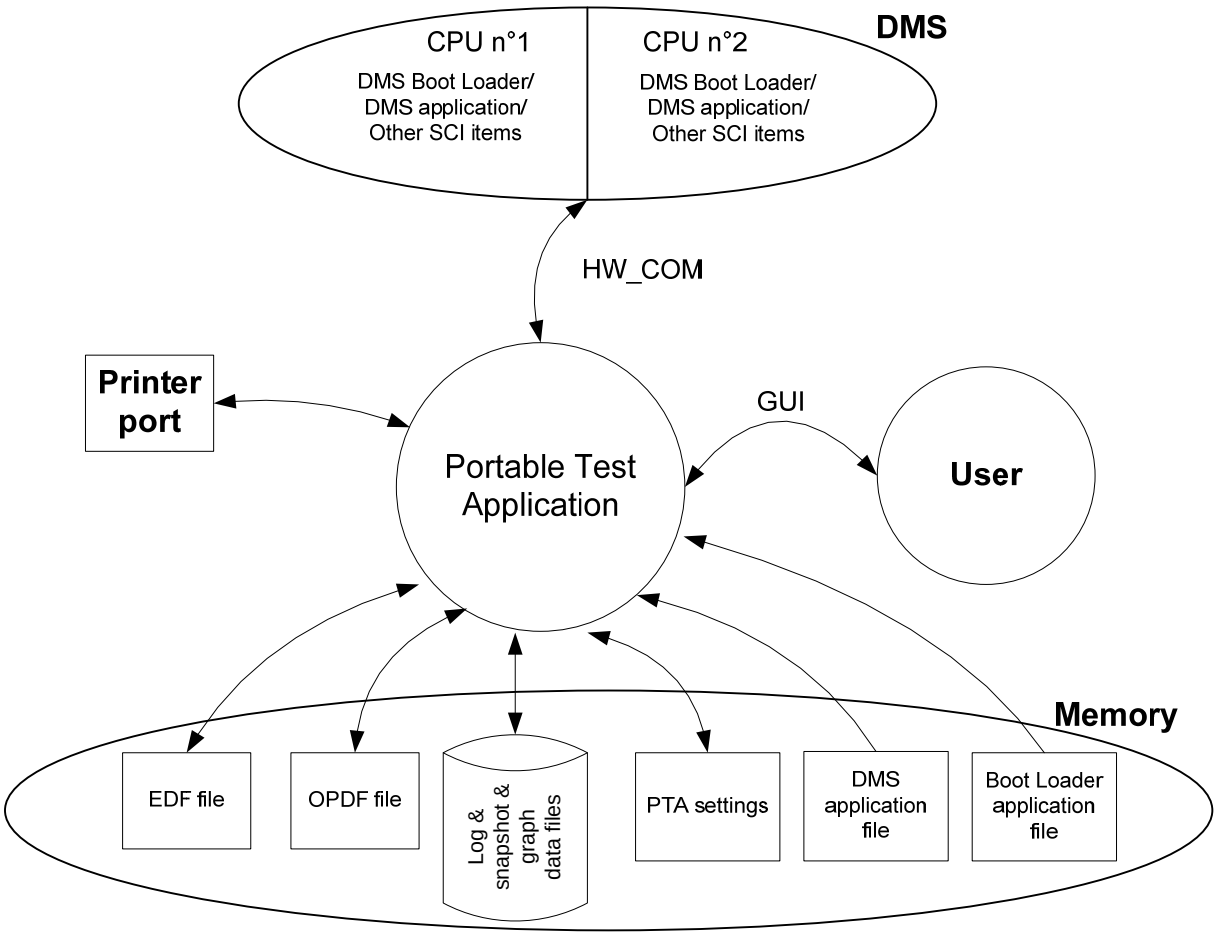
2.1.1 System interfaces

The PTA software communicates with the DMS application or the DMS Boot Loader software to execute user requests. The PTA interfaces to 6 different sub-systems:

- DMS sub-system
- Boot Loader sub-system
- User sub-system
- Memory sub-system
- Printer sub-system

The Figure 2-1 below provides an overview of the PTA system interfaces.

Figure 2-1: PTA software external interface



2.1.2 User interfaces

The PTA communicates with the User through the GUI, a graphical user interface.
The interface operates using both the keyboard and a pointing device (a mouse or a trackball). The commands are entered by activating screen buttons, pull-down menus, or using keyboard shortcuts. It is suggested to design a GUI as close as possible to the interface of the R160 PTA, described in R160 Software User Manual 200310616 Rev. B.

2.1.3 Hardware interfaces

The PTA software is installed on a commercially available notebook computer. It will interface with the following peripherals on the computer :

- Printer port
- Memory
- USB Port or Serial port

2.1.3.1 Memory interface

The memory interface will manage the following files :

- EDF file
- OPDF file
- Log file
- Snapshot file
- Graph data file
- PTA settings
- DMS application file
- Boot Loader application file.

2.1.4 Software interfaces

The PTA software runs in a Microsoft Windows operating system environment. All required libraries are provided in the installation package. No other software is required to use the PTA software. The PTA software will interface with the following software:

- Help file engine

2.1.5 Communications interfaces

The PTA communicates with the target application (when connected) using HW_COM. The term HW_COM may covers several protocols at the same time, meaning that the PTA is capable of communicating via more than one interface (for example USB, Serial or Ethernet), but not at the same time. The commands applicable to this interface are:

- HW_COM.DMS.Message
- HW_COM.BootLoader.Message
- HW_COM.PTA.Message
- HW_COM.PTA.LoginRequest
- HW_COM.DMS.LoginResponse
- HW_COM.PTA.LogoutRequest
- HW_COM.DMS.LogoutResponse
- HW_COM.PTA.RebootRequest
- HW_COM.PTA.EventStatusRequest
- HW_COM.DMS.EventStatusResponse
- HW_COM.PTA.MonitoringRequest
- HW_COM.DMS.MonitoringResponse

- HW_COM.PTA.EventLogFileRequest
- HW_COM.DMS.EventLogFileResponse
- HW_COM.PTA.EraseLogFileRequest
- HW_COM.DMS.EraseLogFileResponse
- HW_COM.PTA.DeleteCommunicationGatewayLogRequest
- HW_COM.DMS.DeleteCommunicationGatewayLogResponse
- HW_COM.PTA.SnapshotFileRequest
- HW_COM.DMS.SnapshotFileResponse
- HW_COM.PTA.EraseSnapshotFileRequest
- HW_COM.DMS.EraseSnapshotFileResponse
- HW_COM.PTA.SelfTestRequest
- HW_COM.DMS.SelfTestResponse
- HW_COM.PTA.DMSApplicationUpgradeRequest
- HW_COM.BootLoader.DMSApplicationUpgradeResponse
- HW_COM.PTA.ReadDMSConfigurationRequest
- HW_COM.DMS.ReadDMSConfigurationResponse
- HW_COM.PTA.WriteDMSConfigurationRequest
- HW_COM.DMS.WriteDMSConfigurationResponse
- HW_COM.PTA.LocalTimeRequest
- HW_COM.DMS.LocalTimeResponse
- HW_COM.PTA.SoftwareVersionsRequest
- HW_COM.DMS.SoftwareVersionsResponse
- HW_COM.PTA.LocationIdentificationRequest
- HW_COM.DMS.LocationIdentificationResponse
- HW_COM.PTA.CarNumberRequest
- HW_COM.DMS.CarNumberResponse
- HW_COM.PTA.CodeStatusRequest
- HW_COM.DMS.CodeStatusResponse
- HW_COM.PTA.MemoryDumpRequest
- HW_COM.DMS.MemoryDumpResponse
- HW_COM.PTA.DoorMovementInformationFileRequest
- HW_COM.DMS.DoorMovementInformationFileResponse
- HW_COM.PTA.EventDescriptionFileRequest
- HW_COM.DMS.EventDescriptionFileResponse
- HW_COM.PTA.OperatingParametersDescriptionFileRequest
- HW_COM.DMS.OperatingParametersDescriptionFileResponse
- HW_COM.PTA.SystemCountersRequest
- HW_COM.DMS.SystemCountersResponse
- HW_COM.PTA.SystemCountersResetRequest
- HW_COM.DMS.SystemCountersResetResponse
- HW_COM.PTA.DoorCommandRequest
- HW_COM.DMS.DoorCommandResponse
- HW_COM.PTA.HeartbeatRequest
- HW_COM.BootLoader.HeartbeatResponse
- HW_COM.DMS.HeartbeatResponse

2.1.6 Memory constraints

The PTA does not have any memory constraints other than those imposed by the capacities of the chosen computer.

2.1.7 User operations

The user operations are explained in section 3.1.4 of this document.

Accessibility to menus and corresponding functions is allowed according to 2 password levels:

Password level	Level	Access to functions
MAINTAINER	0	All functions except : Set PTA properties, Time set, DMS application upgrade, Delete network log file, Delete snapshot, Delete DMS event log file and Reset system counters
ENGINEER	1	All Functions.

Table 2-1: Password levels for PTA

Password level on PTA start-up is given by a internal configuration file. Nevertheless, when the PTA is running, the User can enter the engineer password to access engineer functions.

2.1.8 Site adaptation requirements

There are no site adaptation requirements.

2.2 Product functions

The PTA presents the following functionalities:

- System troubleshooting: door movement information, DMS status, and real-time data monitoring.
- Data retrieval and viewing: event log file, and RTDM files.
- Self test retrieval
- DMS application upgrade
- DMS configuration: open, close timings, and obstruction retry.
- Files and PTA properties

2.3 User characteristics

The primary users of PTAs will be transit authorities service technicians. These users will be trained in servicing particular car systems and in the proper usage of a PTA in this situation. They will be generally familiar with using main operating systems applications. Typical maintenance functions will include fault-log downloading, self-test, fault reset, and non-critical parameter changes. They may be expected to frequently use the PTA.

Secondary users would include management and engineering personnel along with various car builder service and engineering personnel. Engineering functions would include changes to software and certain configurable parameters. The secondary users would be expected to require relatively infrequent usage of a PTA, as compared to the primary user group.

Initial installation of PTA software will be performed by car builder personnel familiar with systems.

2.4 Constraints

- Software development shall conform to the software engineering process CMM-01.
- Software development shall conform to the software engineering policy manual CMM-02.
- The protocol used for the hardware interface between the PTA and the target application shall comply with PTA ICD Document #201111489.

For more design constraints, see section 3.5.

2.5 Assumptions and dependencies

The following lists the factors that can affect the requirements stated in this SRS:

- Change in the features list of the DMS application.
- Change of the OS.
- Change of PTA protocol used for HW_COM defined in PTA ICD Document #201111489 .

2.6 Apportioning of requirements

Not applicable. All requirements are addressed in this document.

3 SPECIFIC REQUIREMENTS

3.1 External interfaces

The external interfaces are shown in Figure 2-1.

3.1.1 HW_COM

The Figure 3-1 depicts the software system logical interfaces and shows how the PTA interfaces with CPU n°1 to CPU n°X applications via HW_COM. As it is possible for a single PTA port to communicate with more than one sub-system, the HW_COM interface protocol has the ability to specify the source destination CPU as shown in Figure 3-1: Software product perspective. The number of supported CPUs is defined on a project per project basis in the SACS document. The commands applicable to this interface are detailed in Table 3-1.

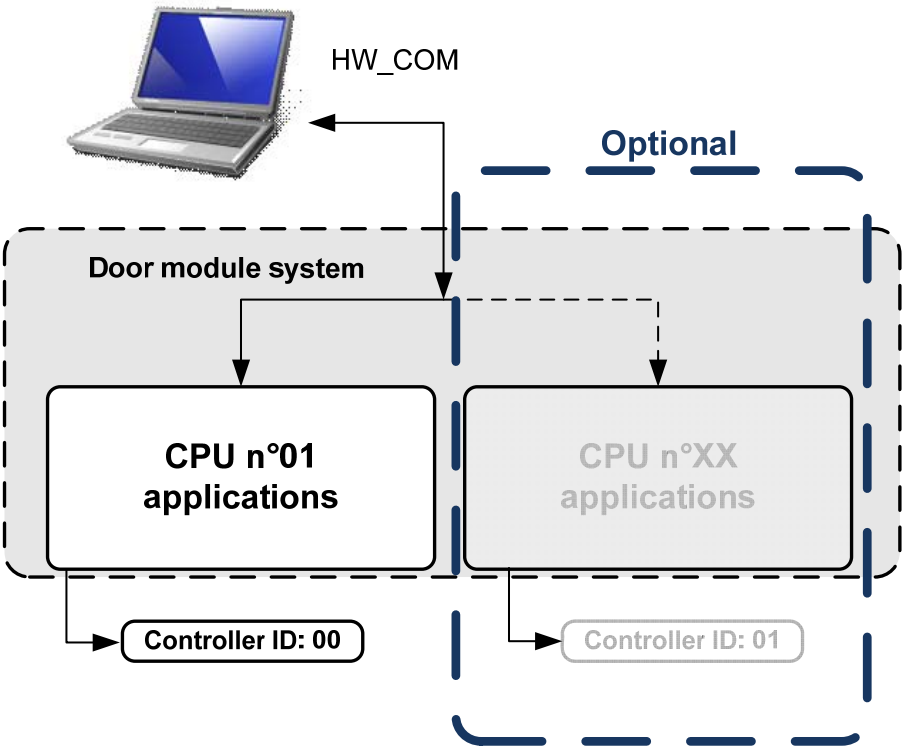


Figure 3-1: Software product perspective

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Table 3-1: Mnemonic of HW_COM elements

Name	Description	Source
HW_COM.DMS.Message	A generic request message sent by the DMS	DMSA
HW_COM.BootLoader.Message	A generic request message sent by the Boot Loader	BLA
HW_COM.PTA.Message	A generic request message sent by the PTA	PTA
HW_COM.PTA.LoginRequest	A request to log into the DMS sent by the PTA	PTA
HW_COM.DMS.LoginResponse	A response of the DMS to HW_COM.PTA.LoginRequest	DMSA
HW_COM.PTA.LogoutRequest	A request to log out of the DMS sent by the PTA	PTA
HW_COM.DMS.LogoutResponse	A response of the DMS to HW_COM.PTA.LogoutRequest	DMSA
HW_COM.PTA.RebootRequest	A request to reboot the DMS sent by the PTA	PTA
HW_COM.PTA.EventStatusRequest	A request for the event status sent by the PTA	PTA
HW_COM.DMS.EventStatusResponse	A response of the DMS to HW_COM.PTA.EventStatusRequest	DMSA
HW_COM.PTA.MonitoringRequest	A request to start the real-time monitoring sent by the PTA	PTA
HW_COM.DMS.MonitoringResponse	A response of the DMS to HW_COM.PTA.MonitoringRequest	DMSA
HW_COM.PTA.EventLogFileRequest	A request for the event log file sent by the PTA	PTA
HW_COM.DMS.EventLogFileResponse	A response of the DMS to HW_COM.PTA.EventLogFileRequest	DMSA
HW_COM.PTA.EraseLogFileRequest	A request to erase the event log files sent by the PTA	PTA
HW_COM.DMS.EraseLogFileResponse	A response of the DMS to HW_COM.PTA.EraseLogFileRequest	DMSA
HW_COM.PTA.DeleteCommunicationGatewayLogRequest	A request to delete the CommunicationGateway log sent by the PTA	PTA
HW_COM.DMS.DeleteCommunicationGatewayLogResponse	A response of the DMS to HW_COM.PTA.DeleteCommunicationGatewayLogRequest	DMSA
HW_COM.PTA.SnapshotFileRequest	A request for the snapshot file sent by the PTA	PTA
HW_COM.DMS.SnapshotFileResponse	A response of the DMS to HW_COM.PTA.SnapshotFileRequest	DMSA
HW_COM.PTA.EraseSnapshotFileRequest	A request to erase the snapshot files sent by the PTA	PTA

Name	Description	Source
HW_COM.DMS.EraseSnapshotFileResponse	A response of the DMS to HW_COM.PTA.EraseSnapshotFileRequest	DMSA
HW_COM.PTA.SelfTestRequest	A request to self test the DMS sent by the PTA	PTA
HW_COM.DMS.SelfTestResponse	A response of the DMS to HW_COM.PTA.SelfTestRequest	DMSA
HW_COM.PTA.DMSApplicationUpgradeRequest	A request to upgrade the DMS application sent by the PTA	PTA
HW_COM.BootLoader.DMSApplicationUpgradeResponse	A response of the DMS to HW_COM.PTA.DMSApplicationUpgradeRequest	DMSA
HW_COM.PTA.ReadDMSConfigurationRequest	A request for reading the DMS configuration file sent by the PTA	PTA
HW_COM.DMS.ReadDMSConfigurationResponse	A response of the DMS to HW_COM.PTA.ReadDMSConfigurationRequest	DMSA
HW_COM.PTA.WriteDMSConfigurationRequest	A request for writing into the DMS configuration file sent by the PTA	PTA
HW_COM.DMS.WriteDMSConfigurationResponse	A response of the DMS to HW_COM.PTA.WriteDMSConfigurationRequest	DMSA
HW_COM.PTA.LocalTimeRequest	A request for the local DMS time sent by the PTA	PTA
HW_COM.DMS.LocalTimeResponse	A response of the DMS to HW_COM.PTA.LocalTimeRequest	DMSA
HW_COM.PTA.SetTimeRequest	A request to set the local DMS time sent by the PTA	PTA
HW_COM.DMS.SetTimeResponse	A response of the DMS to HW_COM.PTA.SetTimeRequest	DMSA
HW_COM.PTA.SoftwareVersionsRequest	A request for the software versions sent by the PTA	PTA
HW_COM.DMS.SoftwareVersionsResponse	A response of the DMS to HW_COM.PTA.SoftwareVersionsRequest	DMSA
HW_COM.PTA.LocationIdentificationRequest	A request for the location identification sent by the PTA	PTA
HW_COM.DMS.LocationIdentificationResponse	A response of the DMS to HW_COM.DMS.LocationIdentificationResponse	DMSA
HW_COM.PTA.CarNumberRequest	A request for the car number sent by the PTA	PTA
HW_COM.DMS.CarNumberResponse	A response of the DMS to HW_COM.PTA.CarNumberRequest	DMSA
HW_COM.PTA.CodeStatusRequest	A request for code status sent by the PTA	PTA
HW_COM.DMS.CodeStatusResponse	A response of the DMS to HW_COM.PTA.CodeStatusRequest	DMSA
HW_COM.PTA.MemoryDumpRequest	A request for the MemoryVolatile.Content sent by the PTA	PTA

Name	Description	Source
HW_COM.DMS.MemoryDumpResponse	A response of the DMS to HW_COM.DMS.MemoryDumpResponse	DMSA
HW_COM.PTA.DoorMovementInformationFileRequest	A request for the door movement information file sent by the PTA	PTA
HW_COM.DMS.DoorMovementInformationFileResponse	A response of the DMS to HW_COM.PTA.DoorMovementInformationFilesRequest	DMSA
HW_COM.PTA.EventDescriptionFileRequest	A request for the event description file sent by the PTA	PTA
HW_COM.DMS.EventDescriptionFileResponse	A response of the DMS to HW_COM.PTA.EventDescriptionFileRequest	DMSA
HW_COM.PTA.OperatingParametersDescriptionFileRequest	A request for the operating parameters description file sent by the PTA	PTA
HW_COM.DMS.OperatingParametersDescriptionFileResponse	A response of the DMS to HW_COM.PTA.OperatingParametersDescriptionFileRequest	DMSA
HW_COM.PTA.SystemCountersRequest	A request for the system counters sent by the PTA	PTA
HW_COM.DMS.SystemCountersResponse	A response of the DMS to HW_COM.PTA.SystemCountersRequest	DMSA
HW_COM.PTA.SystemCountersResetRequest	A request for resetting the system counters sent by the PTA	PTA
HW_COM.DMS.SystemCountersResetResponse	A response of the DMS to HW_COM.PTA.SystemCountersResetRequest	DMSA
HW_COM.PTA.DoorCommandRequest	A response of the DMS to HW_COM.PTA.DoorCommandRequest	PTA
HW_COM.DMS.DoorCommandResponse	A request containing a door command sent by the PTA	DMSA
HW_COM.PTA.HeartbeatRequest	A request of the PTA to verify the presence of the BootLoader.	PTA
HW_COM.BootLoader.HeartbeatResponse	A response to HW_COM.PTA.HeartbeatRequest	BLA
HW_COM.DMS.HeartbeatResponse	A response to HW_COM.PTA.HeartbeatRequest	DMSA

3.1.2 Memory

The memory used by the PTA will be the Hard Disk memory of the computer (see Table 3-2 below).

Table 3-2: Mnemonic of memory elements

Name	Description
Memory.EventLog	Event log file, containing, information about events detected by the system.
Memory.Snapshot	Snapshot file, containing details about all signals valuable for DMS diagnostic.
Memory.DMSApplicationFile	DMS application code
Memory.BootLoaderApplicationFile	Boot Loader application code
Memory.DMSMemoryDump	all memory cells and address lines of the volatile memory
Memory.DoorMovementInformation	Door movement information file, used to store information about latest opening and closing sequence.
Memory.EventDescriptionFile	Event description file, containing events description and parameters.
Memory.OperatingParameterDescriptionFile	Operating parameters description file
Memory.SystemCounters	system counter file containing counters for a number of parameters
Memory.DataFile	A file containing DMS information
Memory.PTASettings	PTA configuration file, containing communication port/speed.

3.1.3 Outputs

The outputs may be software or hardware (see Table 3-3 below).

Table 3-3: Mnemonic of outputs elements

Name	Description
Printer port	Port linked to a printer
Start help file engine	Help file engine : a file browser

3.1.4 VAP4.35 GUI

The graphical user interface used will operate in conjunction with a keyboard and a pointing device. All the information will be displayed on screen through graphical representation and text. The graphical interface is composed of a dialog based interface, made of screen buttons and pull-down menus. The Figure 3-2 is a command tree describing the layout of the GUI. The applicable commands are listed in Table 3-4.

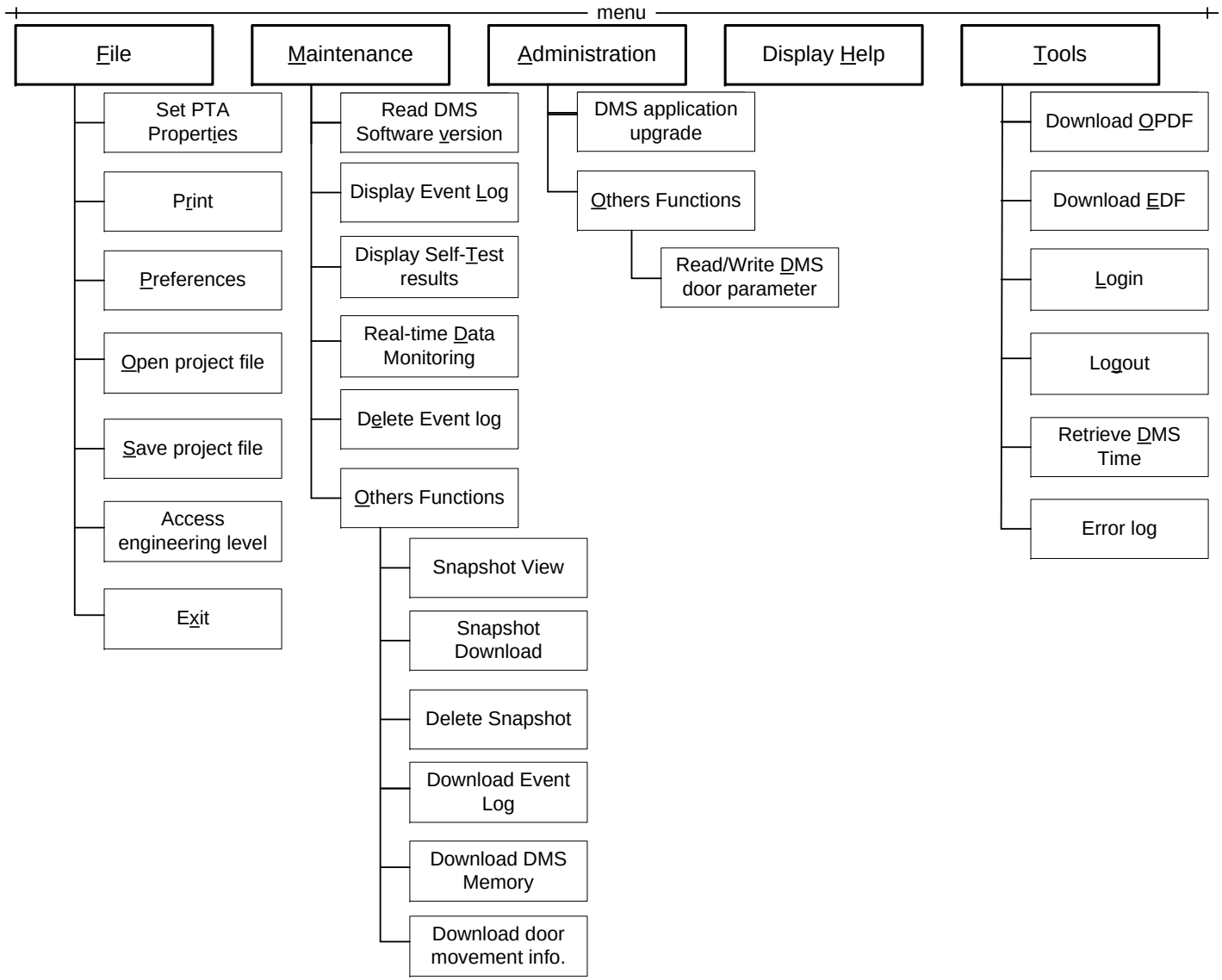

Figure 3-2: GUI Command Tree

Table 3-4: Mnemonic of GUI elements

Name	Description	Source
GUI.User.LoginRequest	A request to log into the DMS sent by the User	User
GUI.PTA.LoginResponse	A response of the PTA to GUI.PTA.LoginRequest	PTA
GUI.User.LogoutRequest	A request to log out of the DMS sent by the User	User
GUI.PTA.LogoutResponse	A response of the PTA to GUI.PTA.LogoutRequest	PTA
GUI.User.RebootRequest	A request to reboot the DMS sent by the User	User
GUI.User.RebootResponse	A response of the PTA to GUI.User.RebootRequest	PTA
GUI.User.EventStatusRequest	A request for the event status sent by the User	User
GUI.PTA.EventStatusResponse	A response of the PTA to GUI.PTA.EventStatusRequest	PTA
GUI.User.MonitoringRequest	A request to start the real-time monitoring sent by the User	User
GUI.PTA.MonitoringResponse	A response of the PTA to GUI.PTA.MonitoringRequest	PTA
GUI.User.EventLogFileRequest	A request for the event log file sent by the User	User
GUI.PTA.EventLogFileResponse	A response of the PTA to GUI.PTA.EventLogFileRequest	PTA
GUI.User.EraseLogFileRequest	A request to erase the event log files sent by the User	User
GUI.PTA.EraseLogFileResponse	A response of the PTA to GUI.PTA.EraseLogFileRequest	PTA
GUI.User.DeleteCommunicationGatewayLogRequest	A request to delete the CommunicationGateway log sent by the User	User
GUI.PTA.DeleteCommunicationGatewayLogResponse	A response of the PTA to GUI.PTA.DeleteCommunicationGatewayLogRequest	PTA
GUI.User.EventLogFileDisplayRequest	A request to display the event log file sent by the User	User
GUI.PTA.EventLogFileDisplay	a response of the PTA to GUI.User.EventLogFileDisplayRequest	PTA
GUI.User.SnapshotFileRequest	A request for the snapshot file sent by the User	User
GUI.PTA.SnapshotFileResponse	A response of the PTA to GUI.PTA.SnapshotFileRequest	PTA
GUI.User.EraseSnapshotFileRequest	A request to erase the snapshot files sent by the User	User

Name	Description	Source
GUI.PTA.EraseSnapshotFileResponse	A response of the PTA to GUI.PTA.EraseSnapshotFileRequest	PTA
GUI.User.SelfTestRequest	A request to self test the DMS sent by the User	User
GUI.PTA.SelfTestResponse	A response of the PTA to GUI.PTA.SelfTestRequest	PTA
GUI.User.DMSApplicationUpgradeRequest	A request to upgrade the DMS application by the User.	User
GUI.PTA.DMSApplicationUpgradeResponse	A response of the PTA to GUI.User.DMSApplicationUpgradeRequest	PTA
GUI.User.ReadDMSConfigurationRequest	A request for reading the DMS configuration file sent by the User	User
GUI.PTA.ReadDMSConfigurationResponse	A response of the PTA to GUI.PTA.ReadDMSConfigurationRequest	PTA
GUI.User.WriteDMSConfigurationRequest	A request for writing into the DMS configuration file sent by the User	User
GUI.PTA.WriteDMSConfigurationResponse	A response of the PTA to GUI.PTA.WriteDMSConfigurationRequest	PTA
GUI.User.LocalTimeRequest	A request for the local DMS time sent by the User	User
GUI.PTA.LocalTimeResponse	A response of the PTA to GUI.PTA.LocalTimeRequest	PTA
GUI.User.SetTimeRequest	A request to set the local DMS time sent by the User	User
GUI.PTA.SetTimeResponse	A response of the PTA to GUI.PTA.SetTimeRequest	PTA
GUI.PTA.DMSLocationIdentificationDisplay	Display of the DMS location identification by the PTA	User
GUI.PTA.DMSCarNumberDisplay	Display of the DMS car number by the PTA	PTA
GUI.User.CodeStatusRequest	A request for code status sent by the User	User
GUI.PTA.CodeStatusResponse	A response of the PTA to GUI.PTA.CodeStatusRequest	PTA
GUI.User.MemoryDumpRequest	A request for the MemoryVolatile.Content sent by the User	User
GUI.PTA.MemoryDumpResponse	A response of the PTA to GUI.DMS.MemoryDumpResponse	PTA
GUI.User.DoorMovementInformationFileRequest	A request for the door movement information file sent by the User	User
GUI.PTA.DoorMovementInformationFileResponse	A response of the PTA to GUI.PTA.DoorMovementInformationFilesRequest	PTA

Name	Description	Source
GUI.User.EventDescriptionFileRequest	A request for the event description file sent by the User	User
GUI.PTA.EventDescriptionFileResponse	A response of the PTA to GUI.PTA.EventDescriptionFileRequest	PTA
GUI.User.OperatingParametersDescriptionFileRequest	A request for the operating parameters description file sent by the User	User
GUI.PTA.OperatingParametersDescriptionFileResponse	A response of the PTA to GUI.PTA.OperatingParametersDescriptionFileRequest	PTA
GUI.User.SystemCountersRequest	A request for the system counters sent by the User	User
GUI.PTA.SystemCountersResponse	A response of the PTA to GUI.User.SystemCountersRequest	PTA
GUI.User.SystemCountersResetRequest	A request for resetting the system counters counter sent by the User	User
GUI.PTA.SystemCountersResetResponse	A response of the PTA to GUI.User.SystemCountersResetRequest	PTA
GUI.User.DoorCommandRequest	A request containing a door command sent by the User	User
GUI.PTA.DoorCommandResponse	A response of the PTA to GUI.PTA.DoorCommandRequest	PTA
GUI.User.LaunchNodeUtilRequest	Request of the User to launch the NodeUtil software	User
GUI.PTA.LaunchNodeUtilResponse	Response of the PTA to GUI.User.LaunchNodeUtilRequest	PTA
GUI.User.OpenFileRequest	Request of the User to open a file	User
GUI.PTA.OpenFileResponse	Response of the PTA to GUI.User.OpenFileRequest	PTA
GUI.User.SaveFileRequest	Request of the User to save a file	User
GUI.PTA.SaveFileResponse	Response of the PTA to GUI.User.SaveFileRequest	PTA
GUI.User.PrintRequest	Request of the User to print a file	User
GUI.PTA.PrintResponse	Response of the PTA to GUI.User.PrintRequest	PTA
GUI.User.ConfigurePTARequest	Request of the User to configure the option of the PTA	User
GUI.PTA.ConfigurePTAResponse	Response of the PTA to GUI.User.ConfigurePTARequest	PTA
GUI.User.HelpRequest	Request of the User to open the help engine	User
GUI.PTA.HelpResponse	Response of the PTA to GUI.User.HelpRequest	PTA
GUI.PTA.ConfirmationRequest	Request of the PTA for a confirmation of the request by the User	User

Name	Description	Source
GUI.User.ConfirmationResponse	Confirmation of the request by the User	PTA
GUI.PTA.ErrorReportDisplay	Error report returned by the PTA after an error occurred	User

3.2 Functions

3.2.1 Online functions

3.2.1.1 VAP4.36 Check incoming and outgoing messages

a) Purpose

The PTA formats every outgoing message before sending it to the DMS or Boot Loader on HW_COM. The PTA also checks every incoming message integrity before processing them.

b) Data

Source	Input
HW_COM	HW_COM.DMS.Message
HW_COM	HW_COM.BootLoader.Message

Destination	Output
HW_COM	HW_COM.PTA.Message
HW_COM	Execute command from incoming message (see following functions for available commands)

c) Processing

1. VAP4.36.1 Before sending HW_COM.PTA.Message, the PTA shall include the integrity check result and the Controller ID value of the CPU it want to address into the message.

2. VAP4.36.2 Upon reception of HW_COM.DMS.Message or HW_COM.BootLoader. Message, the PTA shall verify the integrity of the message content and the validity of the contained command. If found valid, the PTA shall execute the command from the incoming message.

3.2.1.2 VAP4.13 Login

a) Purpose

To avoid other software to accidentally enter and corrupt the DMS Application, a login mechanism is applied. This function allows the DMS and Boot Loader to identify the correct PTA.

b) Data

Source	Input
GUI	GUI.User.LoginRequest
HW_COM	HW_COM.DMS.LoginResponse
HW_COM	HW_COM.BootLoader.LoginResponse

Destination	Output
GUI	GUI.PTA.LoginResponse
HW_COM	HW_COM.PTA.LoginRequest

c) Processing

1. VAP4.13.1 The PTA shall be able to log into CPU n°1 to n°X application at the same time.
2. VAP4.13.2 When the PTA is started, the PTA shall be able to detect if the DMS is present .
3. VAP4.13.4 The PTA shall be able to login to the DMS Application and the DMS Boot Loader .

4. VAP4.13.10 Upon reception of GUI.User.LoginRequest from the User, the PTA shall send HW_COM.PTA.LoginRequest and receive HW_COM.DMS.LoginResponse or HW_COM.BootLoader.LoginResponse in return. The PTA shall then send GUI.PTA.LoginResponse to the User.

3.2.1.3 VAP4.43 Auto-login

a) Purpose

At start-up, the Portable Test Application must check if a connection with a DMS is possible and try to log into it.

b) Data

Source	Input
HW_COM	HW_COM.DMS.LoginResponse
HW_COM	HW_COM.BootLoader.LoginResponse

Destination	Output
HW_COM	HW_COM.PTA.LoginRequest

c) Processing

1. VAP4.43.1 Immediately after start-up, the PTA shall verify if an connection with a DMS is possible and try to log into that DMS by sending HW_COM.PTA.LoginRequest and receive HW_COM.DMS.LoginResponse or HW_COM.BootLoader.LoginResponse in return.

3.2.1.4 VAP4.37 Logout

a) Purpose

The PTA logs out of the DMS or Boot Loader to end the communication.

b) Data

Source	Input
GUI	GUI.User.LogoutRequest
HW_COM	HW_COM.DMS.LogoutResponse
HW_COM	HW_COM.BootLoader.LogoutResponse

Destination	Output
GUI	GUI.PTA.LogoutResponse
HW_COM	HW_COM.PTA.LogoutRequest

c) Processing

1. VAP4.37.1 Upon reception of GUI.User.LogoutRequest from the User, the PTA shall send HW_COM.PTA.LogoutRequest and receive HW_COM.DMS.LogoutResponse or HW_COM.BootLoader.LogoutResponse in return. The PTA shall then send GUI.PTA.LogoutResponse to the User.
2. VAP4.37.2 If the communication is accidentally interrupted, the PTA shall be able to detect the communication error and logout.
3. VAP4.37.3 When the PTA is closed by the user, the PTA notify the DMS to disconnect automatically.

3.2.1.5 VAP4.38 Reboot DMS

a) Purpose
The PTA sends a reboot command to the DMS when the user requests it .

b) Data

Source	Input
GUI	GUI.User.RebootRequest

Destination	Output
GUI	GUI.PTA.RebootResponse
HW_COM	HW_COM.PTA.RebootRequest

c) Processing

1. VAP4.38.1 Upon reception of GUI.User.RebootRequest from the User, the PTA shall send HW_COM.PTA.RebootRequest. The PTA shall then send GUI.PTA.RebootResponse to the User.

3.2.1.6 VAP4.41 Verify connection state

a) Purpose
The purpose of this function is to keep the logged-in state.

b) Data

Source	Input
HW_COM	HW_COM.BootLoader.HeartbeatResponse
HW_COM	HW_COM.DMS.HeartbeatResponse

Destination	Output
HW_COM	HW_COM.PTA.HeartbeatRequest

c) Processing
1. VAP4.41.1 Every 30 seconds, the PTA shall send HW_COM.PTA.HeartbeatRequest to the Boot Loader or DMS and receive respectively HW_COM.BootLoader.HeartbeatResponse or HW_COM.DMS.HeartbeatResponse in answer, to ensure that a working connection is still established between the Boot Loader/DMS and the PTA.

3.2.1.7 VAP4.1 Graphical real-time data monitoring

a) Purpose
The PTA monitor the DMS status upon request.

b) Data

Source	Input
GUI	GUI.User.EventStatusRequest
HW_COM	HW_COM.DMS.EventStatusResponse

Destination	Output
GUI	GUI.PTA.EventStatusResponse
HW_COM	HW_COM.PTA.EventStatusRequest

c) Processing

1. VAP4.1.8 Upon reception of GUI.User.EventStatusRequest from the User, the PTA shall send HW_COM.PTA.EventStatusRequest and receive HW_COM.DMS. EventStatusResponse in return. The PTA shall then send GUI.PTA.EventStatusResponse to the User.
2. VAP4.1.1 The PTA shall display the status for CPU n°1 to CPU n°X DMS applications.
3. VAP4.1.2 When the user opens the DMS status monitor; the PTA software shall keep the status updated in real-time.
4. VAP4.1.3 When the user closes the DMS status monitor, the PTA shall stop updating the status .
5. VAP4.1.4 The DMS status shall include the following information of DMS if applicable to the system: door state, power supply, DMS inputs, DMS outputs, MDC.
6. VAP4.1.7 System status and functional information shall be provided in a graphical format, superimposed over a system schematic or diagrammatic. When a graphic presentation is not practical, system status and functional information shall be provided in english language statements and relevant engineering units, and shall be formatted for simple readability.

3.2.1.8 VAP4.2 Provide text-based real-time monitoring (operating parameters browser)

a) Purpose

The PTA provides a mean to monitor sense point progression in real-time.

b) Data

Source	Input
GUI	GUI.User.MonitoringRequest
HW_COM	HW_COM.DMS.MonitoringResponse

Destination	Output
GUI	GUI.PTA.MonitoringResponse
HW_COM	HW_COM.PTA.MonitoringRequest

c) Processing

1. VAP4.2.1 The PTA shall display operating parameters for CPU n°1 to CPU n°X DMS Applications.
2. VAP4.2.2 Upon reception of GUI.User.MonitoringRequest from user, the PTA shall send HW_COM.PTA.MonitoringRequest to the DMS and receive HW_COM.DMS. MonitoringResponse in response, and then send GUI.PTA.MonitoringResponse to the user.
3. VAP4.2.4 The software shall keep updating the real-time data until the User stop requesting the real-time monitoring.

3.2.1.9 VAP4.3 Download Event log

a) Purpose

The PTA user requests the event log file to be downloaded for inspection/analysis.

b) Data

Source	Input
GUI	GUI.User.EventLogFileRequest
HW_COM	HW_COM.DMS.EventLogFileResponse

Destination	Output
GUI	GUI.PTA.EventLogFileResponse
HW_COM	HW_COM.PTA.EventLogFileRequest
Memory	Memory.EventLog

c) Processing

1. VAP4.3.1 The PTA shall provide the option to download the log file from the CPU n°1 or CPU n°X DMS application.
2. VAP4.3.2 Upon GUI.User.EventLogFileRequest, the PTA software shall send HW_COM.PTA.EventLogFileRequest to the DMS, receive HW_COM.DMS. EventLogFileResponse in response and save the event log it contains into Memory.EventLog. The PTA shall then send GUI.PTA.EventLogFileResponse to the User.
3. VAP4.3.6 When a download error occurs, the error information shall be displayed.
4. VAP4.3.7 Download progression shall appear during event log file transfer.

3.2.1.10 VAP4.6 Display Event log

a) Purpose

The PTA provides display of the event log contents.

b) Data

Source	Input
GUI	GUI.User.EventLogFileDisplayRequest
Memory	Memory.EventLog

Destination	Output
GUI	GUI.PTA.EventLogFileDisplay

c) Processing

1. VAP4.6.1 When GUI.User.EventLogFileDisplayRequest is received by the PTA, it shall send GUI.PTA.EventLogFileDisplay to the User, displaying the event log in the sequence order. .
2. VAP4.6.3 The Log Table shall present the decoded Memory.EventLog database to the user and shall allow sorting by any of its fields.

3.2.1.11 VAP4.4 Delete Event log

a) Purpose

The PTA software erases the DMS event log file, after request and confirmation from the User.

b) Data

Source	Input
GUI	GUI.User.EraseLogFileRequest
HW_COM	HW_COM.DMS.EraseLogFileResponse
GUI	GUI.User.ConfirmationResponse

Destination	Output
GUI	GUI.PTA.EraseLogFileResponse
HW_COM	HW_COM.PTA.EraseLogFileRequest
GUI	GUI.PTA.ConfirmationRequest

c) Processing

1. VAP4.4.1 The PTA shall provide the option to delete the log file of the CPU n°1 or CPU n°X DMS application.
2. VAP4.4.2 Upon reception of GUI.User.EraseLogFileRequest from User, the PTA shall send GUI.PTA.ConfirmationRequest to the User, and get GUI.User.ConfirmationResponse in return. The PTA shall then send HW_COM.PTA.EraseLogFileRequest to the DMS and receive HW_COM.DMS.EraseLogFileResponse when the DMS deleted the event log. The PTA shall then send GUI.PTA.EraseLogFileResponse to the User.

3.2.1.12 VAP4.5 Delete network log file

a) Purpose

The PTA software erases the CommunicationGateway log file, after request and confirmation from the User.

b) Data

Source	Input
GUI	GUI.User.DeleteCommunicationGatewayLogRequest
GUI	GUI.User.ConfirmationResponse
HW_COM	HW_COM.DMS.DeleteCommunicationGatewayLogResponse

Destination	Output
GUI	GUI.PTA.ConfirmationRequest
HW_COM	HW_COM.PTA.DeleteCommunicationGatewayLogRequest
GUI	GUI.PTA.DeleteCommunicationGatewayLogResponse

c) Processing

1. VAP4.5.1 The PTA shall provide the option to delete the log file of the CommunicationGateway application of any DMS.

2. VAP4.5.3 Upon reception GUI.User.DeleteCommunicationGatewayLogRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation and receive GUI.User.ConfirmationResponse in return. The PTA shall then send HW_COM.PTA.DeleteCommunicationGatewayLogRequest to the DMS and receive HW_COM.DMS.DeleteCommunicationGatewayLogResponse upon deletion of the CommunicationGateway log. The PTA shall then send GUI.PTA.DeleteCommunicationGatewayLogResponse to the user to notify the success of the deletion.

3.2.1.13 VAP4.7 Download RTDM snapshot file

a) Purpose

The PTA downloads the RTDM Snapshot file for system analysis.

b) Data

Source	Input
GUI	GUI.User.SnapshotFileRequest
HW_COM	HW_COM.DMS.SnapshotFileResponse

Destination	Output
HW_COM	HW_COM.PTA.SnapshotFileRequest
Memory	Memory.Snapshot
GUI	GUI.PTA.SnapshotFileResponse

c) Processing

1. VAP4.7.1 The PTA shall provide the option to download the RTDM snapshot file from CPU n°1 or CPU n°X DMS application.
2. VAP4.7.2 Upon reception of GUI.User.SnapshotFileRequest from user, the PTA shall send HW_COM.PTA.SnapshotFileRequest, receive HW_COM.DMS.SnapshotFileResponse in return and save its content into Memory.Snapshot. The PTA shall then send GUI.PTA.SnapshotFileResponse to the user.
3. VAP4.7.4 Upon reception, Memory.Snapshot shall be uncompressed, decoded according to OPDF data format for each operating parameter and added to snapshot list. The PTA software tool shall be able to view all snapshot contents.
4. VAP4.7.5 When download error occurs, the error information shall be displayed to the user.
5. VAP4.7.6 Download progression shall appear during snapshot file transfer.

3.2.1.14 VAP4.8 Delete RTDM snapshot

a) Purpose

The PTA software deletes the RTDM snapshot file after request and confirmation from the User.

b) Data

Source	Input
GUI	GUI.User.EraseSnapshotFileRequest
HW_COM	HW_COM.DMS.EraseSnapshotFileRespon se
GUI	GUI.User.ConfirmationResponse

Destination	Output
HW_COM	HW_COM.PTA.EraseSnapshotFileRequest
GUI	GUI.PTA.ConfirmationRequest
GUI	GUI.PTA.EraseSnapshotFileResponse

c) Processing

1. VAP4.8.1 The PTA shall provide the option to delete snapshot file to CPU n°01 or CPU n°X DMS application.
2. VAP4.8.2 Upon reception of GUI.User.EraseSnapshotFileRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation, and receive GUI.User.ConfirmationResponse in return. The PTA shall then send HW_COM.PTA.EraseSnapshotFileRequest to the DMS and receive HW_COM.DMS.EraseSnapshotFileResponse after the snapshot file has been deleted. The PTA shall then send GUI.PTA.EraseSnapshotFileResponse to notify the User of the success of the operation

3.2.1.15 VAP4.9 Display DMS self test results on demand

a) Purpose

The PTA sends self test request to the DMS, and displays the DMS self test results on screen. The results are obtained through the DMS self test periodic procedure.

b) Data

Source	Input
GUI	GUI.User.SelfTestRequest
HW_COM	HW_COM.DMS.SelfTestResponse

Destination	Output
HW_COM	HW_COM.PTA.SelfTestRequest
GUI	GUI.PTA.SelfTestResponse

c) Processing

1. VAP4.9.1 The PTA shall be able to request self test results to CPU n°1 to CPU n°X DMS applications .
2. VAP4.9.3 Upon reception of GUI.User.SelfTestRequest from the User, , the PTA shall send HW_COM.PTA.SelfTestRequest to the DMS and receive HW_COM.DMS.SelfTestResponse containing the self-test results. The PTA shall then send GUI.PTA.SelfTestResponse to the User, displaying the results on the screen.

3.2.1.16 VAP4.10 DMS application upgrade

a) Purpose

The PTA upgrades the DMS application when connected to the Boot Loader. The PTA cannot upgrade the DMS application if connected to it.

b) Data

Source	Input
GUI	GUI.User.DMSApplicationUpgradeRequest
GUI	GUI.User.ConfirmationResponse
HW_COM	HW_COM.BootLoader.DMSApplicationUpgradeResponse
Memory	Memory.DMSApplicationFile

Destination	Output
GUI	GUI.PTA.ConfirmationRequest
HW_COM	HW_COM.PTA.DMSApplicationUpgradeRequest
GUI	GUI.PTA.DMSApplicationUpgradeResponse

c) Processing

1. VAP4.10.1 The PTA shall provide the option to upgrade CPU n°1 or CPU n°X DMS application.
2. VAP4.10.2 Upon reception of GUI.User.DMSApplicationUpgradeRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation and receive GUI.User.ConfirmationResponse in answer. The PTA shall then send HW_COM.PTA.DMSApplicationUpgradeRequest, containing the application code, to the Boot Loader and receive HW_COM.BootLoader.DMSApplicationUpgradeResponse upon completion of the application upgrade. The PTA shall then send GUI.PTA.DMSApplicationUpgradeResponse to notify the User of the success of the operation.
3. VAP4.10.5 Errors compromising the integrity of the DMS software shall interrupt the DMS upgrade and be reported to the user.
4. VAP4.10.6 Download progression shall appear during the DMS application file transfer.

3.2.1.17 VAP4.44 Boot Loader application upgrade

a) Purpose

The purpose of this function is to allow upgrade of the Boot Loader application.

b) Data

Source	Input
Memory	Memory.BootLoaderApplicationFile

Destination	Output
HW_COM	Boot Loader application

c) Processing

1. VAP4.44.1 The PTA shall perform upgrade of the Boot Loader application saved as Memory.BootLoaderApplicationFile through the HW_COM interface. The implementation of this function is specific to the CPU being used in the DMS and shall be define in the design phase.

3.2.1.18 VAP4.39 Read DMS parameters

a) Purpose

The PTA displays to the User a number of parameters impacting the functioning of the Door System, allowing the user to see the current parameters before modifying them.

b) Data

Source	Input
GUI	GUI.User.ReadDMSConfigurationRequest
HW_COM	HW_COM.DMS.ReadDMSConfigurationResponse

Destination	Output
HW_COM	HW_COM.PTA.ReadDMSConfigurationRequest
GUI	GUI.PTA.ReadDMSConfigurationResponse

c) Processing

1. VAP4.39.1 The PTA shall provide the option to configure CPU n°1 or CPU n°X DMS application.

2. VAP4.39.2 Upon reception of GUI.User.ReadDMSConfigurationRequest from the User, the PTA shall send HW_COM.PTA.ReadDMSConfigurationRequest to the DMS and receive GUI.PTA.ReadDMSConfigurationResponse. The PTA shall then send GUI.PTA.ReadDMSConfigurationResponse to the User, displaying the current parameters.

3.2.1.19 VAP4.11 Write DMS door parameters

a) Purpose

The PTA provides function that allows the User to modify a number of parameters impacting the functioning of the Door Module System.

b) Data

Source	Input
GUI	GUI.User.WriteDMSConfigurationRequest
HW_COM	HW_COM.DMS.WriteDMSConfigurationResponse

Destination	Output
GUI	GUI.PTA.WriteDMSConfigurationResponse
HW_COM	HW_COM.PTA.WriteDMSConfigurationRequest

c) Processing

1. VAP4.11.1 The PTA shall provide the option to configure CPU n°1 or CPU n°X DMS application.
2. VAP4.11.3 Upon reception of GUI.User.WriteDMSConfigurationRequest from the User, the PTA shall send HW_COM.PTA.WriteDMSConfigurationRequest to the DMS and receive GUI.PTA.WriteDMSConfigurationResponse upon modification of the parameters by the DMS. The PTA shall then send GUI.PTA.WriteDMSConfigurationResponse to the User..

3.2.1.20 VAP4.12 Detect data communication error

a) Purpose

The PTA is able to detect and try to correct the error during data transmission.

b) Data

Source	Input
HW_COM	HW_COM.DMS.Message
HW_COM	HW_COM.BootLoader.Message

Destination	Output
GUI	GUI.PTA.ErrorReportDisplay
HW_COM	HW_COM.PTA.Message

c) Processing

1. VAP4.12.2 The PTA shall be able to detect error occurred in communication, such as packet integrity error and time out, when receiving HW_COM.DMS.Message or HW_COM. BootLoader.Message or when sending HW_COM.PTA.Message.
2. VAP4.12.3 A mechanism shall be implemented to avoid non-end communication waiting.
3. VAP4.12.4 If PTA fails to recover from a communication error, the error information shall be reported to the user through GUI.PTA.ErrorReportDisplay.

3.2.1.21 VAP4.14 Retrieve DMS time/date

a) Purpose
The PTA retrieves the DMS time.

b) Data

Source	Input
GUI	GUI.User.LocalTimeRequest
HW_COM	HW_COM.DMS.LocalTimeResponse

Destination	Output
HW_COM	HW_COM.PTA.LocalTimeRequest
GUI	GUI.PTA.LocalTimeResponse

c) Processing
1. VAP4.14.1 The PTA shall provide the option to retrieve the time from CPU n°1 to n°X DMS application.
2. VAP4.14.2 Upon reception of GUI.User.LocalTimeRequest, the PTA shall send HW_COM.PTA.LocalTimeRequest to the DMS and receive HW_COM.DMS. LocalTimeResponse in return, containing the time of the CPU. The PTA shall then send GUI.PTA.LocalTimeResponse to the User, displaying the DMS time.
3. VAP4.14.3 The DMS date (mm/dd/yyyy) and time (hh:mm:ss AM/PM) shall be displayed in local time.

3.2.1.22 VAP4.42 Set DMS time/date

a) Purpose
The PTA set the DMS time and date.

b) Data

Source	Input
GUI	GUI.User.SetTimeRequest
HW_COM	HW_COM.DMS.SetTimeResponse

Destination	Output
HW_COM	HW_COM.PTA.SetTimeRequest
GUI	GUI.PTA.SetTimeResponse

c) Processing
1. VAP4.42.1 The PTA shall provide the option to set the time from CPU n°1 to n°x DMS application.
2. VAP4.42.2 Upon reception of GUI.User.SetTimeRequest, the PTA shall send HW_COM.PTA. SetTimeRequest to the DMS and receive HW_COM.DMS. SetTimeResponse in return, containing the time of the CPU. The PTA shall then send GUI.PTA. SetTimeResponse to the User, displaying the DMS time.

3.2.1.23 VAP4.15 Read DMS software version

a) Purpose
The PTA requests the SCI items software versions.

b) Data

Source	Input
HW_COM	HW_COM.DMS.SoftwareVersionsResponse
HW_COM	HW_COM.BootLoader.SoftwareVersionsResponse

Destination	Output
HW_COM	HW_COM.PTA.SoftwareVersionsRequest

c) Processing
1. VAP4.15.1 The PTA shall provide the option to retrieve the software version from CPU n°1 or CPU n°X application.
2. VAP4.15.2 After login, the PTA shall send HW_COM.PTA.SoftwareVersionsRequest to the DMS and receive HW_COM.DMS.SoftwareVersionsResponse or HW_COM.BootLoader.SoftwareVersionsResponse in answer, containing the DMS SCI items software versions.
3. VAP4.15.4 When available, the DMS software versions shall be displayed .
4. VAP4.15.5 The DMS software versions shall include the DMS application version, the Boot Loader version and , if applicable, CommunicationGateway application version.

3.2.1.24 VAP4.16 Read door location identification and car number

a) Purpose
The PTA requests the door location identification in the car and the car number.

b) Data

Source	Input
HW_COM	HW_COM.DMS.LocationIdentificationResponse
HW_COM	HW_COM.DMS.CarNumberResponse

Destination	Output
HW_COM	HW_COM.PTA.LocationIdentificationRequest
HW_COM	HW_COM.PTA.CarNumberRequest
GUI	GUI.PTA.DMSLocationIdentificationDisplay
GUI	GUI.PTA.DMSCarNumberDisplay

c) Processing
1. VAP4.16.1 The PTA shall provide the option to retrieve the Location identification and car number from CPU n°1 or CPU n°X application.
2. VAP4.16.2 Automatically after login into the DMS, the PTA shall send HW_COM.PTA.LocationIdentificationRequest and HW_COM.PTA.CarNumberRequest to the DMS and receive respectively HW_COM.DMS.LocationIdentificationResponse and HW_COM.DMS.CarNumberResponse from the DMS, containing respectively the location identification and the car number.
3. VAP4.16.4 Upon reception of the information, the PTA shall send GUI.PTA.DMSLocationIdentificationDisplay to the User, displaying the door location identification.
4. VAP4.16.5 Upon reception of the information, the PTA shall send GUI.PTA.DMSCarNumberDisplay to the User, displaying the door car number.

3.2.1.25 VAP4.22 Read DMS code status

a) Purpose

The PTA reads current DMS code status upon request from the User.

b) Data

Source	Input
GUI	GUI.User.CodeStatusRequest
HW_COM	HW_COM.DMS.CodeStatusResponse
HW_COM	HW_COM.BootLoader.CodeStatusRespons e

Destination	Output
HW_COM	HW_COM.PTA.CodeStatusRequest
GUI	GUI.PTA.CodeStatusResponse

c) Processing

1. VAP4.22.1 The PTA shall provide the option to retrieve the code status from CPU n°1 or CPU n°X application.
2. VAP4.22.2 Upon reception of GUI.User.CodeStatusRequest from the User, the PTA shall send HW_COM.PTA.CodeStatusRequest to the DMS and receive HW_COM.DMS. CodeStatusResponse or HW_COM.BootLoader.CodeStatusResponse in return, containing the DMS code status. The PTA shall then send GUI.PTA. CodeStatusResponse to the User, displaying the DMS code status on screen.
3. VAP4.22.4 The DMS code status shall include the DMS application, the DMS Boot Loader application, and, if applicable, the CommunicationGateway application code status.

3.2.1.26 VAP4.23 Download DMS memory

a) Purpose

The PTA provides the function to download and view the DMS volatile memory contents. The beginning and the end of the memory zone to download can be defined by the User.

b) Data

Source	Input
GUI	GUI.User.MemoryDumpRequest
HW_COM	HW_COM.DMS.MemoryDumpResponse

Destination	Output
GUI	GUI.PTA.MemoryDumpResponse
HW_COM	HW_COM.PTA.MemoryDumpRequest
Memory	Memory.DMSMemoryDump

c) Processing

1. VAP4.23.1 The PTA shall provide the option to retrieve the volatile memory content from the CPU n°1 or CPU n°X application.
2. VAP4.23.6 The PTA shall provide the option to set the starting address and the end address of the memory zone to download
3. VAP4.23.2 Upon reception of GUI.User.MemoryDumpRequest from the User, the PTA shall send HW_COM.PTA.MemoryDumpRequest to the DMS, receive HW_COM.DMS .MemoryDumpResponse in return and save its content into Memory.DMSMemoryDump . The PTA shall then send GUI.PTA.MemoryDumpResponse to the User, displaying the memory dump result on screen.

4. VAP4.23.4 The DMS volatile memory contents shall be displayed in both Hex and ASCII format.
5. VAP4.23.5 Download progression shall appear during DMS memory file transfer.

3.2.1.27 VAP4.25 Download door movement information (graphs)

a) Purpose

The PTA user requests the door movement information file to be downloaded for inspection /analysis.

b) Data

Source	Input
GUI	GUI.User.DoorMovementInformationFileRequest
HW_COM	HW_COM.DMS.DoorMovementInformationFileResponse

Destination	Output
HW_COM	HW_COM.PTA.DoorMovementInformationFileRequest
Memory	Memory.DoorMovementInformation
GUI	GUI.PTA.DoorMovementInformationFileResponse

c) Processing

1. VAP4.25.1 The PTA shall provide the option to download the door movement information file from CPU n°1 and CPU n°X DMS application.
2. VAP4.25.2 Upon reception of GUI.User.DoorMovementInformationFileRequest from the User, the PTA shall send HW_COM.PTA.DoorMovementInformationFileRequest to the DMS, receive HW_COM.DMS.DoorMovementInformationFileResponse in return and save its content into Memory.DoorMovementInformation. The PTA shall then send GUI.PTA.DoorMovementInformationFileResponse to the User, displaying the data on screen.
3. VAP4.25.5 The PTA shall present door movement information graphically to the user.
4. VAP4.25.6 When a download error occurs the Error Information shall be displayed.
5. VAP4.25.7 Download progression shall appear during graph file transfer.

3.2.1.28 VAP4.26 Download Event Description File (EDF)

a) Purpose

The PTA provides the function to update EDF from the DMS.

b) Data

Source	Input
GUI	GUI.User.EventDescriptionFileRequest
HW_COM	HW_COM.DMS.EventDescriptionFileResponse

Destination	Output
HW_COM	HW_COM.PTA.EventDescriptionFileRequest
Memory	Memory.EventDescriptionFile
GUI	GUI.PTA.EventDescriptionFileResponse

c) Processing

1. VAP4.26.1 Upon reception of GUI.User.EventDescriptionFileRequest from the User, the PTA shall send HW_COM.PTA.EventDescriptionFileRequest to the DMS, receive HW_COM.DMS.EventDescriptionFileResponse in return and save its content into Memory.EventDescriptionFile. The PTA shall then send GUI.PTA.EventDescriptionFileResponse to the User, displaying the data on screen.
2. VAP4.26.3 The DMS EDF Contents shall be able to be browsed by the user.
3. VAP4.26.4 Download progression shall appear during EDF file transfer.

3.2.1.29 VAP4.27 Download Operating Parameter Description File (OPDF)

a) Purpose

The PTA provides the function to update OPDF from the DMS.

b) Data

Source	Input
GUI	GUI.User.OperatingParametersDescriptionFileRequest
HW_COM	HW_COM.DMS.OperatingParametersDescriptionFileResponse

Destination	Output
HW_COM	HW_COM.PTA.OperatingParametersDescriptionFileRequest
Memory	Memory.OperatingParameterDescriptionFile
GUI	GUI.PTA.OperatingParametersDescriptionFileResponse

c) Processing

1. VAP4.27.1 Upon reception of GUI.User.OperatingParametersDescriptionFileRequest from the User, the PTA shall send HW_COM.PTA.OperatingParametersDescriptionFileRequest to the DMS, receive HW_COM.DMS.OperatingParametersDescriptionFileResponse in return and save its content into Memory.OperatingParameterDescriptionFile. The PTA shall then send GUI.PTA.OperatingParametersDescriptionFileResponse to the User, displaying the data on screen.
2. VAP4.27.3 The DMS OPDF Contents shall be able to be browsed by the user.
3. VAP4.27.4 Download progression shall appear during OPDF file transfer.

3.2.1.30 VAP4.32 Download system counters

a) Purpose

The PTA provides the function to retrieve the DMS System Counters.

b) Data

Source	Input
GUI	GUI.User.SystemCountersRequest
HW_COM	HW_COM.DMS.SystemCountersResponse

Destination	Output
GUI	GUI.PTA.SystemCountersResponse
HW_COM	HW_COM.PTA.SystemCountersRequest
Memory	Memory.SystemCounters

c) Processing

1. VAP4.32.5 Upon reception of GUI.User.SystemCountersRequest from the User, the PTA shall send HW_COM.PTA.SystemCountersRequest to the DMS, receive HW_COM.DMS.SystemCountersResponse in return and save its content into Memory.SystemCounters. The PTA shall then send GUI.PTA.SystemCountersResponse to the User, displaying the data on screen.
2. VAP4.32.8 The DMS System Counters Contents shall be displayed to the user.
3. VAP4.32.6 Download progression shall appear during System counters file transfer. The access to other PTA functions should be disabled until the data transfer is completed or cancelled.

3.2.1.31 VAP4.40 Reset system counters

a) Purpose

The PTA shall send a request to reset the system counters upon request from the User.

b) Data

Source	Input
GUI	GUI.User.SystemCountersResetRequest
HW_COM	HW_COM.DMS.SystemCountersResetResponse
GUI	GUI.User.ConfirmationResponse

Destination	Output
GUI	GUI.PTA.SystemCountersResetResponse
HW_COM	HW_COM.PTA.SystemCountersResetRequest
GUI	GUI.PTA.ConfirmationRequest

c) Processing

1. VAP4.40.1 Upon reception of GUI.User.SystemCountersResetRequest from the User, the PTA shall send GUI.PTA.ConfirmationRequest for confirmation and receive GUI.User.ConfirmationResponse in return from the User. the PTA shall then send HW_COM.PTA.SystemCountersResetRequest to the DMS and receive HW_COM.DMS.SystemCountersResetResponse in return. The PTA shall then send GUI.PTA.SystemCountersResetResponse to the User, notifying him of the success of the reset .

3.2.1.32 VAP4.17 Open project file

a) Purpose

The PTA opens project files which contains log and snapshot data saved on the PTA computer hard disk for review and analysis.

b) Data

Source	Input
GUI	GUI.User.OpenFileRequest
Memory	Memory.DataFile

Destination	Output
GUI	GUI.PTA.OpenFileResponse

c) Processing

1. VAP4.17.1 Upon reception of GUI.User.OpenFileRequest from the User, the PTA shall send GUI.PTA.OpenFileResponse to the user, open the selected file from Memory. DataFile and display it on screen.
2. VAP4.17.2 Bad file name, type and file data shall be detected and reported to the user.

3.2.1.33 VAP4.18 Save project file

a) Purpose

The PTA should save the current Project including log and snapshot data in DMS memory to the PTA computer hard disk for future use.

b) Data

Source	Input
GUI	GUI.User.SaveFileRequest

Destination	Output
GUI	GUI.PTA.SaveFileResponse
Memory	Memory.DataFile

c) Processing

1. VAP4.18.1 Upon reception of GUI.User.SaveFileRequest from the User, the PTA shall send GUI.PTA.SaveFileResponse to the user and save the selected file from Memory.DataFile.

3.2.1.34 VAP4.19 Print

a) Purpose

The PTA prints the DMS Log and Snapshot.

b) Data

Source	Input
GUI	GUI.User.PrintRequest

Destination	Output
GUI	GUI.PTA.PrintResponse
Printer port	Memory.DataFile

c) Processing

1. VAP4.19.1 Upon reception of GUI.User.PrintRequest from the User, the PTA shall send GUI.PTA.PrintResponse to the User and print the selected file (Memory.DataFile) in predefined format.

3.2.1.35 VAP4.20 Set PTA properties

a) Purpose

The PTA software sets up the working data directory and communication parameters.

b) Data

Source	Input
GUI	GUI.User.ConfigurePTARequest

Destination	Output
GUI	GUI.PTA.ConfigurePTAResponse
Memory	Memory.PTASettings

c) Processing

1. VAP4.20.1 The PTA shall provide interfaces to allow user to change current working directory, to set up Hardware Interface(s) and to define login options .
2. VAP4.20.2 Upon reception of GUI.User.ConfigurePTARequest from the User, the PTA shall send GUI.PTA.ConfigurePTAResponse and save the changes into Memory.PTASettings.
3. VAP4.20.3 Each time the PTA starts, it shall read PTA Settings from the system registry and set up the PTA.
4. VAP4.20.4 The setup error shall be reported to the user.

3.2.1.36 VAP4.21 Display help

a) Purpose

The PTA gets online help for user.

b) Data

Source	Input
GUI	GUI.User.HelpRequest

Destination	Output
GUI	GUI.PTA.HelpResponse
Output	Start help file engine

c) Processing

1. VAP4.21.1 Upon reception of GUI.User.HelpRequest from the User, the PTA shall send GUI.PTA.HelpResponse to the User and start the Help File engine to load help into memory .
2. VAP4.21.2 User shall be able to use this help engine to browse the help file.

3.2.1.37 VAP4.24 Access Engineering level

a) Purpose

The PTA provides the function to prevent unauthorized update to the DMS.

b) Data

Source	Input
GUI	GUI.User.DMSApplicationUpgradeRequest
GUI	GUI.User.WriteDMSConfigurationRequest
GUI	GUI.User.EraseSnapshotFileRequest
Input	Password

Destination	Output
Various	Execute command from incoming message (see Source list for possible messages)

c) Processing

1. VAP4.24.2 The PTA user shall be prohibited from performing the following requests unless the PTA detects the presence of the password with the proper Firm and User code: GUI.User.DMSApplicationUpgradeRequest, GUI.User.WriteDMSConfigurationRequest, GUI.User.EraseSnapshotFileRequest.

3.3 VAP4.28 Performance requirements

VAP4.28.2 The time required to upload the entire software for the door system, including time to replace firmware embodied in CPUs, shall be no more than one hour.

3.4 Logical database requirements

There is no third party external logical database used. The PTA uses the .NET framework libraries to manage data.

3.5 Design constraints

3.5.1 Standards compliance

The PTA's design and code is not required to comply to any standard.

3.5.2 Hardware limitations

The PTA must run on a standard commercially available personal computer provided by the car builder. The hardware specifications for the computer are to be announced by the car builder.

3.5.3 Software limitations

The PTA must run on a Microsoft Windows operating system environment.

3.6 VAP4.29 Software system attributes

3.6.1 Reliability

There is no specific reliability requirement for this software.

3.6.2 Availability

There is no availability requirement for this software.

3.6.3 Security

See 3.2.1.37.

3.6.4 Maintainability

There is no maintainability requirement for this software.

3.6.5 Portability

The PTA software is only required to run in a Microsoft Windows operating system environment. All necessary libraries and API will be installed with the PTA installation.